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Inventor(s)

Gannan; Dylan Jacob

Bait Station

Abstract

A bait station for pest control, the bait station comprising: an enclosure defining an access chamber and a bait chamber, said access chamber including at least one entry port for permitting entry of a pest into said enclosure, said bait chamber configured to hold therein bait material harmful to the pest; and a load sensing module arranged in the bait chamber for sensing changes in weight of bait material positioned in said bait chamber and operable to output signal indicative of sensed changes in weight of the bait material and transmit said output signal to a remote location.

Inventors: Gannan; Dylan Jacob (Rockhampton, AU)

Applicant: PestSense Holdings Pty Limited (Brisbane, AU)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/030,718, titled “Bait Station,” filed on Apr. 6, 2023, and currently pending; U.S. application Ser. No. 18/030,718 is a U.S. Nat’l Stage Application of Patent Cooperation Treaty App. PCT/AU2020/051077, titled “Bait Station,” filed on Oct. 7, 2020, and now expired. The entire contents of U.S. application Ser. No. 18/030,718 and Patent Cooperation Treaty App. PCT/AU2020/051077, including but not limited to their originally-filed claims, are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates to bait stations for combating pests such as mice and, rats that include electronic systems and methods for detecting the presence and monitoring the activity of such pests.

BACKGROUND

[0003] Any references to methods, apparatus or documents of the prior art are not to be taken as constituting any evidence or admission that they formed, or form part of the common general knowledge.

[0004] Small animal pests, such as mice and rats, are typically controlled by the strategic placement of traps and/or poisons. Such traps and/or poisons are placed in areas of known or suspected pest activity. In order for such control methods to be used efficiently and effectively, it is important to have accurate information concerning pest activity, including information on bait consumption and the patterns of bait consumption. Such information may be used, for example, to minimize the number of traps required to control a pest population and replenish bait stations with bait and possibly even schedule replenishment of bait stations based on consumption patterns.

[0005] Electronic detection and counting devices may be used for monitoring pest activity. An example of such a device is described in U.S. Pat. No. 5,040,326 to Van Dijnsen, et al. This patent describes a system for detecting pests which includes a luring box having an entrance aperture sized to let a small rodent enter the box. A detector or sensor is mounted in the box near the entrance aperture, and is activated by the presence of a rodent in the box. Such previously known electronic pest monitoring devices are limited in their usefulness both by the limited pest activity information which they provide and in their lack of ease of use. Such devices typically provide only a count which is related to the number of pests entering a selected location in which the monitoring device is positioned.

[0006] What is desired, therefore, is an improved electronic pest monitoring system which provides sufficiently detailed pest activity information to provide effective and efficient pest control, and which is both easy to deploy and use.

SUMMARY OF INVENTION

[0007] In an aspect, the invention provides a bait station for pest control, the bait station comprising:

[0008] an enclosure defining an access chamber and a bait chamber, said access chamber including at least one entry port for permitting entry of a pest into said enclosure, said bait chamber configured to hold therein bait material and a load sensing module arranged in the bait chamber for sensing changes in weight of bait material positioned in said bait chamber and operable to output signal indicative of sensed changes in weight of the bait material and transmit said output signal to a remote location.

[0009] In an embodiment, the bait station further comprises a detector module associated with the access chamber for detecting presence of a pest in said access chamber and operative to output a signal indicative of presence or otherwise of a pest.

[0010] In an embodiment, the detector module is operatively coupled with the load sensing module to activate the load sensing module to sense changes in the weight of the bait material in response to detecting presence of the pest in the access chamber.

[0011] In an embodiment, the load sensing module comprises one or more load cells arranged relative to a platform portion of the bait chamber, each load cell being coupled with bait holding members for sensing changes in weight of the bait material.

[0012] In an embodiment, the bait holding members comprise upright posts coupled with the load cells positioned along an underside of the bait chamber.

[0013] In an embodiment, walls defining an interior volume of the bait station comprise one or more apertures for allowing the bait holding members to pass through and be coupled with each load cell located thereunder.

[0014] In an embodiment, a plurality of said bait holding members are coupled to a single cell.

[0015] In another embodiment, the bait holding member extends across the internal volume of the bait chamber in a substantially horizontal bait holding orientation, the bait holding member being elongate and extending between two ends such that each end is supported by an in-use upper portion of a supporting clip adapted to be fastened to bait holding member wherein the lower portion of the supporting clip is operatively coupled with the load sensing module.

[0016] In an embodiment, the load sensing module comprises a load sensing cell positioned on a load cell supporting bar extending at an inclination angle (a) relative to a base surface of the enclosure, the load supporting bar and the load sensing cell being configured for sensing changes in weight of bait material positioned on the bait holding member when the bait station is positioned in a plurality of in-use orientations.

[0017] In an embodiment, the inclination angle is less than 90 degrees and more preferably in the range of 15 degrees and 75 degrees and more preferably at an angle of substantially 45 degrees.

[0018] In an embodiment, the bait chamber is defined by at least two laterally spaced apart upstanding walls extending from the base surface and wherein a respective load sensing module is positioned adjacent each of the upstanding walls.

[0019] In an embodiment, each respective load sensing module is positioned in a partially enclosed cavity defined by a corresponding upstanding wall and an outer wall of the bait station.

[0020] In an embodiment, the load cell engaging bar is mounted on one or more mounting locations along an internal surface of the outer wall defining said cavity.

[0021] In an embodiment, the load cell engaging bar is mounted on two spaced apart mounting locations to orient the load cell supporting bar at the inclination angle (a).

[0022] In an embodiment, the load cell supporting bar extends transversely relative to a plane of the upstanding wall to arrange the load cell engaging bar and the load sensing cell in a configuration for sensing changes in weight of bait material positioned on the bait holding member when the bait station is positioned in a plurality of in-use orientations.

[0023] In an embodiment, one or both of the upstanding walls further comprise a slot located along an in-use edge of the upstanding walls to allow the bait holding member to pass through when supported on the load cell during use.

[0024] In an embodiment, the load sensing module is located outside an internal space of the bait chamber or the access chamber to prevent damage to electrical componentry by pests.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] Preferred features, embodiments and variations of the invention may be discerned from the following Detailed Description which provides sufficient information for those skilled in the art to perform the invention. The Detailed Description is not to be regarded as limiting the scope of the

preceding Summary of the Invention in any way. The Detailed Description will make reference to a number of drawings as follows:

[0026] FIG. 1 is a top perspective view of a bait station **100** in accordance with an embodiment.

[0027] FIG. 2 is a top view of the bait station **100**.

[0028] FIG. 3 is a bottom view of the bait station **100**.

[0029] FIG. 4 is a schematic view of the detection module **160** and the load sensing module **140**.

[0030] FIG. 5 is a top perspective view of a bait station **200** in accordance with an embodiment.

[0031] FIG. 6 is an underside view of the bait stations **200**.

[0032] FIG. 7 is an enlarged view of the load sensing module **240** used in the bait station **200**.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0033] FIGS. 1 to 4 illustrate a first embodiment of the bait station **100** in accordance with an embodiment of the invention. In relation to the figures, FIGS. 1 and 2 illustrate top views of a bait station **100** in an open state (whereby the top cover or lid has been removed). The bait station **100** comprises a housing **110**, which is formed from lid portion (not shown) and a base portion **104**. The lid portion and base portion **104**, in one or more embodiments, may be connected by a hinge arrangement or any other arrangement provided in the region of adjacent longitudinal edges of the lid portion and base portion **104**. With such a configuration, the lid and base portions may combine to resemble a “clam-shell”. The lid may be provided with access ports (not shown) to allow the pest to enter the access chamber **120** that provides an enclosed internal space bound by convergent lateral walls **122**. The access chamber **120** is formed continuously with and leads into a bait chamber **130**. The bait chamber **130** is structured to hold bait material that is harmful to the pest. The invention is not limited in any manner by the nature of bait material used and all types of bait material may be used without departing from the scope of the presently described embodiment.

[0034] The bait chamber **130** includes a load sensing module **140** arranged in the bait chamber for sensing changes in weight of bait material positioned in the bait chamber **130**. The load sensing module is operable to output signal indicative of sensed changes in weight of the bait material and transmit the output signal to a remote location. FIG. 4 illustrates an exemplary and non-limiting embodiment of the load sensing module **140**. The load sensing module **140** may be conveniently provided on a PCB circuit board and may include a plurality of load cells (also known as transducers) **142A** and **142B** (generally denoted by **142**). Each load cell **142** is operatively coupled with bait holding posts **132** that extend orthogonally relative to the base **104** of the housing enclosure **110**. The load sensing module **140** is preferably located along an underside portion of the base **104** (to minimise exposure to water, moisture and pest faeces) and the four bait holding posts **132** extend upwardly from the base **104** to allow bait material to be loaded on said bait holding posts **132**.

[0035] The load cells **142** may be in electrical communication with an actuation switch **143** to activate or de-activate the load cells **142**. During use, the load cells **142** are arranged to sense any changes in weight of the bait material loaded or supported on the bait holding posts. Signal data from the load cells **142** may be communicated to a remotely located server or computing device (not shown) via a trans-receiver unit **145** which forms a part of the load sensing module **140**. Output signals from the load cells **142** may be transmitted via the transreceiver unit **147** to provide an indication of any changes in weight of the bait material loaded on the bait holding posts. In some embodiments, the trans-receiver unit **147** may be operable to be in communication with an on-board microprocessor **149** to undertake processing of the signals before the processed signals are communicated via the trans-receiver unit **147**. The load cells **142** may undertake measurements at a pre-determined sampling rate. The output signal received at the remote location may be processed to determine changes in the weight of bait material during specified periods of time. It would be understood that a power source such as but not limited to a battery (including a rechargeable battery) may be provided to power the load sensing module **140**.

[0036] At the entrance portion of the access chamber **120**, a detector module **160** is provided for

detecting presence of a pest as the pest enters the access chamber **120**. The detector module **160** is operative to output a signal indicative of presence of a pest in the access chamber **120**. In the preferred embodiment, the detector module **160** may include emitter detector pairs (electro-optic or infrared devices) denoted by **162** and **164** such that as the pest enters the access chamber **120**, it would be expected to pass through the electromagnetic radiation beam extending between one of the emitter/detector pairs **162** and **164**. Each time, the electromagnetic beam is broken by the entry and/or exit of the pest, an output signal indicative of the presence of the pest may be generated and transmitted to a remote location. In some embodiments, the detector module may save the number of breaks detected by the detector module **160** and this number may be saved and transmitted on a periodic basis for conserving energy in the system. In the preferred embodiment, the detector module **160** may be coupled with the load sensing module to activate the load sensing module **140** (via the actuation switch) when the pest enters the access chamber **120** to undertake weight measurements for the bait materials during periods when the pest is present in the access chamber. In other embodiments, the load sensing module **140** may undertake weight measurements at pre-determined intervals independent of any triggering. The detector module **160** may also generate an output signal associate with the presence of the pest and communicate information related to the presence of the pest via the trans-receiving unit **147** of the load sensing module **140**. The combination of the aforementioned detection module **160** and the load sensing module **140** provides data related to bait consumption patterns for various pests and helps monitor and schedule bait replenishment for the bait stations. In at least some optional embodiments, the usage of the load sensing module **140** only during periods when the pest is present within the enclosure **110** of the bait station **100** alleviates the requirement to keep the load sensing module **140** constantly activated. The arrangement of the load sensing module **140** and the detector module **160** also allows any changes of weight of bait material to be co-related with pest movement detected by the detector module **160**.

[0037] Any data associated with the detector module **160** and/or the load sensing module **140** may be transmitted to a remote location via one or more wireless technologies such as LORA (Long Range) or short range wireless technologies including Wi-Fi and CATM1(cellular).

[0038] It is important to appreciate that the load sensing module **140** either alone or in combination with the detector module **160** may be retrofitted into existing bait stations to upgrade existing bait stations.

[0039] Referring to FIGS. **5** to **7**, a second embodiment of the bait station **200** has been illustrated. Like reference numerals denote like features which have been previously described. Unlike the previously described embodiment, the bait station **200** comprises a large and centrally located bait chamber **230** that with an access opening **220** that allows the pest into the bait chamber **230**. The detector module **160** may be located at the access opening **220** to detect the presence of a pest entering the bait chamber **230**. The bait holding member **232** in an in-use position extends across the internal volume of the bait chamber **230** in a substantially horizontal bait holding orientation. The bait holding member **232** is elongate and extends between two ends such that each end is supported by an in-use upper portion of a supporting clip **234** adapted to be fastened to bait holding member **232**. The supporting clip **234** is operatively coupled with a respective load sensing module **240**. It is important to appreciate that the load sensing module **240** also comprises electronic load cells (similar to load cells **142** described earlier) that are adapted to measure any changes in weight of bait material loaded on the bait holding member **232**.

[0040] The load sensing module **240** comprises a load cell engaging bar **236** (which is integrally formed with the supporting clip **234**) that extends at an inclination angle (which may be generally denoted by a) relative to the horizontal base surface of the bait chamber **230**. The load cell engaging bar **236** engages with load sensing cells **142** positioned thereunder. The load cell engaging bar **236** is configured for sensing changes in weight of bait material positioned on the bait holding member **232** when the bait station **200** is positioned in a plurality of in-use orientations. In

the preferred embodiment, the inclination angle α is substantially 45 degrees. However, this angle is not limiting and may be varied to other angles between 0 and 90 degrees without departing from the intended scope of the invention.

[0041] The bait chamber **230** is defined by the two laterally spaced apart upstanding walls **222** extending from the base surface and the respective load sensing modules **240** are positioned adjacent each of the upstanding walls. Specifically, each load sensing module **240** is enclosed positioned in a partially enclosed cavity defined by a corresponding upstanding wall **222** and an outer wall **224** of the bait station **200**. The load cell engaging bar **236** is mounted on two spaced apart mounting members **225** (best shown in FIG. 6) along an internal surface of the outer wall **224** the upstanding wall **222**. The mounting member **225** are arranged in a manner such that the load cell engaging member **234** extends transversely relative to a plane of the upstanding wall **222** to arrange the load cell engaging bar **236** and the load sensing cell in a configuration for sensing changes in weight of bait material positioned on the bait holding member when the bait station is positioned in a plurality of in-use orientations. In the preferred embodiment, the load cells are positioned on the mounting members **225** and the load cell engaging bar **236** is mounted thereon. Advantageously, both upstanding walls **222** include slots **227** located along an in-use edge of the upstanding walls to allow the bait holding member **232** to pass through when supported on the load cell during use.

[0042] The angled configuration of the load cell engaging bar **236** in combination with load sensing module **240** and the bait holding member **232** allows the bait station **200** to be used in a plurality of orientations. For example, in a narrow space, the bait station **200** may be positioned to align the base surface **220** of the bait station **200** to be substantially vertical. In other scenarios, the bait station may be positioned on undulating or sloping surfaces at an angle. The novel configuration of the load cell engaging member **236** allows changes in weight of bait material (as loaded on the bait holding member **232**) to be sensed irrespective of the orientation. The transverse or angled orientation of the load cell engaging member **236** ensures that in all of the plurality of orientations, a vertical weight component of the bait material acts on the load cells positioned below the transversely oriented load cell engaging member **236**.

[0043] Additionally, an accelerometer (such as an electronic gyroscopic module) may also be coupled with the load sensing module to allow measurement or sensing of the in-use orientation of the bait station **200**. Such information would allow users to, more accurately calculate, the real weight of the bait material loaded on the bait holding member **232**.

[0044] In compliance with the statute, the invention has been described in language more or less specific to structural or methodical features. The term “comprises” and its variations, such as “comprising” and “comprised of” is used throughout in an inclusive sense and not to the exclusion of any additional features.

[0045] It is to be understood that the invention is not limited to specific features shown or described since the means herein described comprises preferred forms of putting the invention into effect.

[0046] The invention is, therefore, claimed in any of its forms or modifications within the proper scope of the appended claims appropriately interpreted by those skilled in the art.

ADDITIONAL EXAMPLE EMBODIMENTS

[0047] Example 1: A bait station for pest control, the bait station comprising: (i) an enclosure defining an access chamber and a bait chamber, said access chamber including at least one entry port for permitting entry of a pest into said enclosure, said bait chamber configured to hold therein bait material; and (ii) a load sensing module arranged in the bait chamber for sensing changes in weight of bait material positioned in said bait chamber and operable to output signal indicative of sensed changes in weight of the bait material and transmit said output signal to a remote location.

[0048] Example 2: A bait station in accordance with Example 1 further comprising a detector module associated with the access chamber for detecting presence of a pest in said access chamber

and operative to output a signal indicative of presence or otherwise of a pest.

[0049] Example 2: A bait station in accordance with Example 2 wherein the detector module is operatively coupled with the load sensing module to activate the load sensing module to sense changes in the weight of the bait material in response to detecting presence of the pest in the access chamber.

[0050] Example 4: A bait station in accordance with any one of the preceding Examples wherein the load sensing module one or more load cells arranged relative to a platform portion of the bait chamber, each load cell being coupled with bait holding members for sensing changes in weight of the bait material.

[0051] Example 5: A bait station in accordance with Example 4 wherein the bait holding members comprise upright posts coupled with the load cells positioned along an underside of the bait chamber.

[0052] Example 6: A bait station in accordance with Example 4 or Example 5 wherein walls defining an interior volume of the bait station comprise one or more apertures for allowing the bait holding members to pass through and be coupled with each load cell located thereunder.

[0053] Example 7: A bait station in accordance with Example 5 or Example 6 wherein a plurality of said bait holding members are coupled to a single cell.

[0054] Example 8: A bait station in accordance with any one of the preceding Examples 1 to 7 wherein the bait holding member extends across the internal volume of the bait chamber in a substantially horizontal bait holding orientation, the bait holding member being elongate and extending between two ends such that each end is supported by an in-use upper portion of a supporting clip adapted to be fastened to bait holding member wherein the lower portion of the supporting clip is operatively coupled with the load sensing module.

[0055] Example 9: A bait station in accordance with Example 8 wherein the load sensing module comprises a load sensing cell positioned on a load cell supporting bar extending at an inclination angle (α) relative to a base surface of the enclosure, the load supporting bar and the load sensing cell being configured for sensing changes in weight of bait material positioned on the bait holding member when the bait station is positioned in a plurality of in-use orientations.

[0056] Example 10: A bait station in accordance with Example 9 wherein the inclination angle is less than 90 degrees and more preferably in the range of 15 degrees and 75 degrees and more preferably at an angle of substantially 45 degrees.

[0057] Example 11: A bait station in accordance with any one of Examples 8 to 10 wherein the bait chamber is defined by at least two laterally spaced apart upstanding walls extending from the base surface and wherein a respective load sensing module is positioned adjacent each of the upstanding walls.

[0058] Example 12: A bait station in accordance with Example 11 wherein each respective load sensing module is positioned in a partially enclosed cavity defined by a corresponding upstanding wall and an outer wall of the bait station.

[0059] Example 13: A bait station in accordance with Example 12 when dependent upon Example 9 wherein the load cell engaging bar is mounted on one or more mounting locations along an internal surface of the outer wall defining said cavity.

[0060] Example 14: A bait station in accordance with Example 12 wherein the load cell engaging bar is mounted on two spaced apart mounting locations to orient the load cell supporting bar at the inclination angle (α).

[0061] Example 15: A bait station in accordance with Example 14 wherein during use, the load cell engaging bar extends transversely relative to a plane of the upstanding wall to arrange the load supporting bar and the load sensing cell in a configuration for sensing changes in weight of bait material positioned on the bait holding member when the bait station is positioned in a plurality of in-use orientations.

[0062] Example 16: A bait station in accordance with Example 14 or Example 15 wherein one or

both of the upstanding walls further comprise a slot located along an in-use edge of the upstanding walls to allow the bait holding member to pass through when supported on the load cell during use. [0063] Example 17: A bait station in accordance with any one of the preceding Examples wherein the load sensing module is located outside an internal space of the bait chamber or the access chamber to prevent damage to electrical componentry by pests.

Claims

1. A bait station for pest control, the bait station comprising: an enclosure defining an access chamber and a bait chamber, wherein the access chamber includes at least one entry port configured to permit entry of a pest into the enclosure, and wherein the bait chamber is configured to hold therein bait material; and a load sensing module, wherein: (i) the load sensing module is configured to sense changes in weight of bait material positioned in said bait chamber, and (ii) the load sensing module is configured to output an output signal indicative of the sensed changes in the weight of the bait material, and (iii) the load sensing module or the bait station is configured to transmit said output signal to a remote location; and wherein: (a) the load sensing module is arranged in the bait chamber, or (b) the load sensing module is located outside an internal space of the bait chamber or the access chamber to prevent damage to electrical componentry by pests.
2. The bait station in accordance with claim 1, further comprising a detector module associated with the access chamber, wherein (i) the detector module is configured to detect presence of a pest in the access chamber, and (ii) the detector module is configured to output a signal indicative of presence or otherwise of a pest.
3. The bait station in accordance with claim 2, wherein the detector module is operatively coupled with the load sensing module and is configured to activate the load sensing module to sense changes in the weight of the bait material in response to detecting presence of the pest in the access chamber.
4. The bait station in accordance with claim 1, wherein the load sensing module comprises one or more load cells arranged relative to a platform portion of the bait chamber, wherein each load cell is coupled with one or more bait holding members configured to sense changes in weight of the bait material.
5. The bait station in accordance with claim 4, wherein the one or more bait holding members comprise upright posts coupled with the one or more load cells positioned along an underside of the bait chamber.
6. The bait station in accordance with claim 4, wherein walls defining an interior volume of the bait station comprise one or more apertures for allowing the one or more bait holding members to pass through and to be coupled with each load cell located thereunder.
7. The bait station in accordance with claim 5, wherein a plurality of the bait holding members are coupled to a single load cell.
8. The bait station in accordance with claim 1, wherein a bait holding member extends across an internal volume of the bait chamber in a substantially horizontal bait holding orientation, wherein the bait holding member is elongate and extends between two ends such that each end of the bait holding member is supported by an in-use upper portion of a supporting clip, wherein the supporting clip is configured to support or to fasten to the bait holding member, wherein a lower portion of the supporting clip is operatively coupled with the load sensing module.
9. The bait station in accordance with claim 8, wherein the load sensing module comprises a load sensing cell positioned in engagement with a load cell engaging bar extending at an inclination angle (α) of less than 90 degrees relative to a base surface of the enclosure, and wherein the load cell engaging bar and the load sensing cell are configured to sense changes in weight of bait material positioned on the bait holding member when the bait station is positioned in any of a plurality of in-use orientations.

- 10.** The bait station in accordance with claim 9, wherein the load sensing cell engages with and is positioned under the load cell engaging bar.
- 11.** The bait station in accordance with claim 9, wherein the inclination angle is in a range of from 15 degrees to 75 degrees.
- 12.** The bait station in accordance with claim 8, wherein the bait chamber is defined by at least two laterally spaced apart upstanding walls extending from a base surface, and wherein a respective load sensing module is positioned adjacent each of the upstanding walls.
- 13.** The bait station in accordance with claim 12, wherein each respective load sensing module is positioned in a partially enclosed cavity defined by a corresponding upstanding wall and an outer wall of the bait station.
- 14.** The bait station in accordance with claim 9, wherein the bait chamber is defined by at least two laterally spaced apart upstanding walls extending from the base surface, and wherein a respective load sensing module is positioned adjacent each of the upstanding walls; wherein each respective load sensing module is positioned in a partially enclosed cavity defined by a corresponding upstanding wall and an outer wall of the bait station; and wherein the load cell engaging bar is mounted on one or more mounting locations along an internal surface of the outer wall defining said cavity.
- 15.** The bait station in accordance with claim 9, wherein the bait chamber is defined by at least two laterally spaced apart upstanding walls extending from the base surface, and wherein a respective load sensing module is positioned adjacent each of the upstanding walls; wherein each respective load sensing module is positioned in a partially enclosed cavity defined by a corresponding upstanding wall and an outer wall of the bait station; and wherein the load cell engaging bar is mounted on two spaced apart mounting locations to orient the load cell engaging bar at the inclination angle (α).
- 16.** The bait station in accordance with claim 15, wherein, during use, the load cell engaging bar extends transversely relative to a plane of the upstanding wall to arrange the load cell engaging bar and the load sensing cell in a configuration to sense changes in weight of bait material positioned on the bait holding member when the bait station is positioned in any of a plurality of in-use orientations.
- 17.** The bait station in accordance with claim 15, wherein one or both of the upstanding walls further comprise a slot located along an in-use edge of the upstanding walls to allow the bait holding member to pass through when supported on the load cell during use.
- 18.** The bait station in accordance with claim 1, wherein the load sensing module is located outside an internal space of the bait chamber or the access chamber to prevent damage to electrical componentry by pests.
- 19.** The bait station in accordance with claim 1, wherein the load sensing module is configured to undertake weight measurements at pre-determined intervals independently of any triggering.
- 20.** The bait station in accordance with claim 1, wherein the load sensing module comprises a plurality of load cells, and wherein the load cells are configured to undertake weight measurements at a pre-determined sampling rate.
- 21.** The bait station in accordance with claim 1, wherein the bait station comprises a trans-receiver unit which is configured to transmit, to a remotely located server or to a remotely located computing device, the output signal indicative of the sensed changes in the weight of the bait material.
- 22.** A bait station comprising: an enclosure comprising: (i) an access chamber having at least one entry port for pest ingress; and (ii) a bait chamber configured to hold bait material; a weight monitoring system comprising a platform portion in the bait chamber; a load sensing module comprising one or more load cells arranged relative to the platform portion; an elongate bait holding member extending horizontally across the bait chamber between opposing ends of the bait station; a pair of supporting clips, wherein each supporting clip comprises: (i) an upper portion

supporting one of the opposing ends of the elongate bait holding member; and (ii) a lower portion operatively coupled to the load sensing module; a load cell engaging bar arranged at an acute angle relative to a base surface of the enclosure; and a load sensing cell operatively coupled to the load cell engaging bar, wherein the load sensing module is configured to: (i) detect changes in weight of bait material supported by the elongate bait holding member when the bait station is oriented in any of a plurality of different positions, and (ii) transmit signals indicating the detected weight changes to a remote location.

23. The bait station of claim 22, further comprising a detector module associated with the access chamber and configured to detect presence of a pest, wherein: the detector module is operatively coupled to the load sensing module; the load sensing module is configured to sense changes in weight of the bait material in response to the detector module detecting presence of the pest.

24. The bait station of claim 22, wherein: the load cell engaging bar is mounted on two spaced apart mounting locations arranged to orient the load cell engaging bar at an inclination angle between 15 degrees and 75 degrees relative to the base surface; the load cell engaging bar extends transversely relative to a plane of an upstanding wall of the bait chamber; and the load sensing module is positioned in a partially enclosed cavity defined between the upstanding wall and an outer wall of the bait station.

25. The bait station of claim 22, wherein: the load sensing module is configured to undertake weight measurements at predetermined intervals independently of any triggering; the bait chamber is defined by two laterally spaced apart upstanding walls extending from the base surface; each upstanding wall comprises a slot located along an edge configured to allow the elongate bait holding member to pass through; and a respective load sensing module is positioned adjacent to each of the upstanding walls in a partially enclosed cavity.
